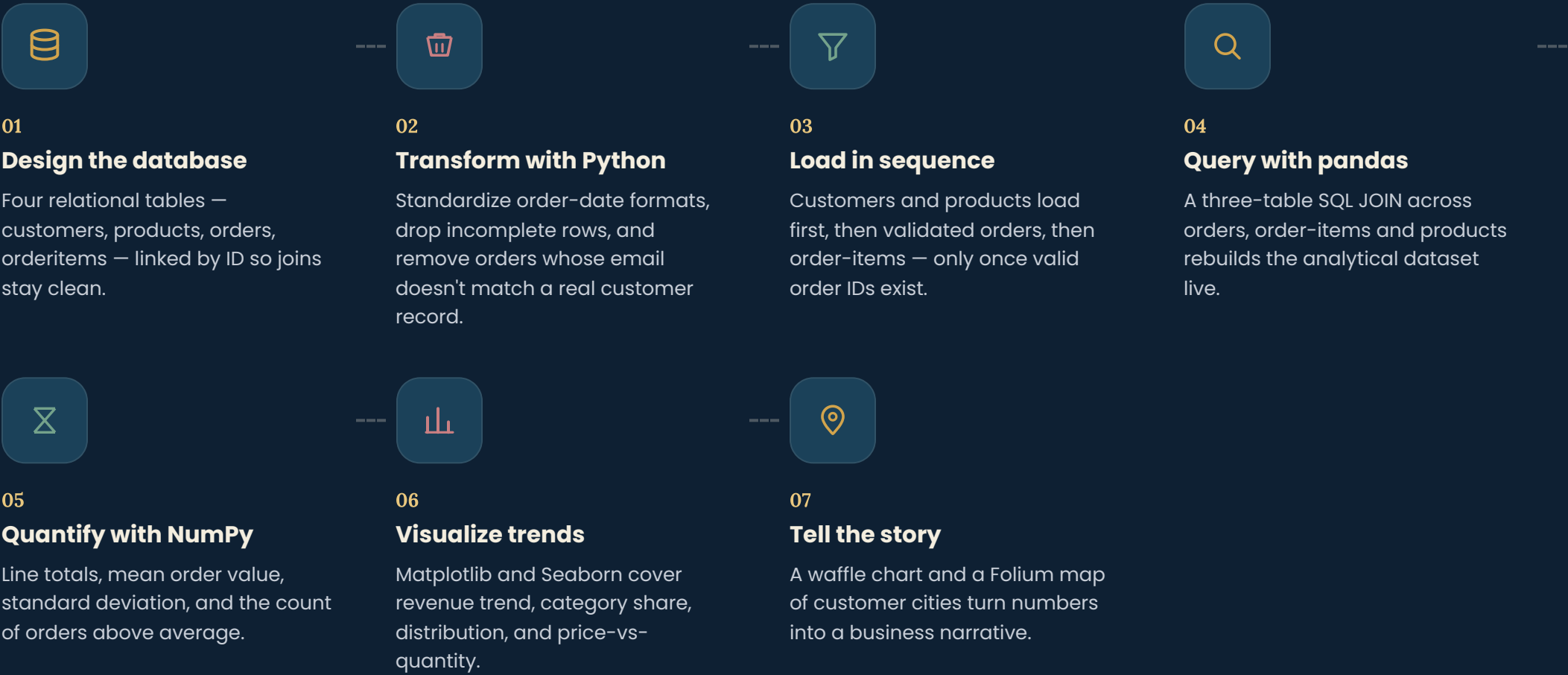


BookNook Books.

A small bookstore's messy order data, turned into a live relational database and a set of business-ready insights — built end to end with **PostgreSQL, pandas, NumPy, and four visualization libraries** to practice the full data journey: extract, transform, load, analyze, and tell the story.

The pipeline, seven stages



What the data showed

\$111.95

Total analyzed revenue
across 5 order-lines, Sept 1–3

58%

of revenue from Fiction
the clear leading category

\$61.47

Best single day
Sept 3 — nearly double day one

Islamabad

top customer city
5 customers on the map

From numbers to narrative

● Revenue share, at a glance

A PyWaffle chart makes Fiction's 58% share immediately readable — one square, one intuitive unit.

● Where the customers are

A Folium map sizes each city by customer count — Islamabad and Lahore lead the footprint.

Built with

- PostgreSQL
- Python
- pandas
- NumPy
- Matplotlib
- Seaborn
- PyWaffle
- Folium

Three things worth remembering

Sequencing is the pipeline

Order-items can only load once valid order IDs exist — loading order matters as much as the data itself.

Know your std dev formula

pandas defaults to a sample std dev (ddof=1); NumPy defaults to population (ddof=0) — same data, different number.

Reproducibility needs a reset

Seeded IDs assume a clean table — a fresh reset before rerunning the ETL avoids silent mismatches.